Geostatistical Refinement of GIS Classification Results

Evaluating and Correcting Spatial Outliers via Kriging-based Hypothesis Testing

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Master's Thesis Defense
2026

Presentation Outline

Background: Pixel-based Classification in GIS

Context: Remote Sensing and Expectations Maximization (EM) Clustering

- High-resolution data demands automated, pixel-wise Land Use/Land Cover (LULC) classification or clustering.
- The EM clustering algorithm was used in the data used in this study, and five clustering results were given.

Tobler's 1st Law

"Everything is related to everything else, but near things are more related than distant things." [4]

The Flaw in Expectation-Maximization Algorithm: EM Clustering treat every single pixel as an independent sample. this directly violates Tobler's First Law of Geography. Natural landscapes are continuous and spatially autocorrelated.

Problem Statement: The 'Salt-and-Pepper' Effect

The Phenomenon: This happens when individual pixels, or small, fragmented groups, stand out against their background. Their labels, as assigned by the EM algorithm, clash with the surrounding data, creating a noisy, "speckled" appearance that doesn't match the environment. [1].

Consequences in GIS

- Degraded LULC decision-making.
- Inaccurate spatial area estimation.
- Hindered landscape ecology metrics.

Limitations of Heuristic Filters

- **Majority Filter:** Purely geometric, destroys thin linear features (e.g., roads, rivers) [3].
- **Markov Random Fields (MRF):** Tends to oversmooth genuine geographic boundaries.
- **Object-Based Image Analysis (OBIA):** Relies heavily on subjective parameters.

Research Objectives & Pipeline

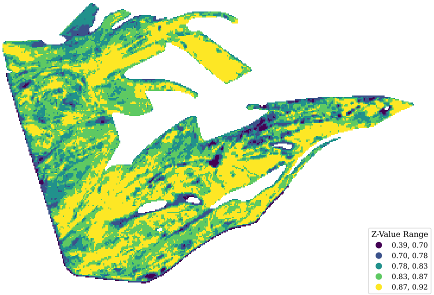
The Core Objective

Correct isolated labels so they make sense in the context of the surrounding physical landscape (Z).

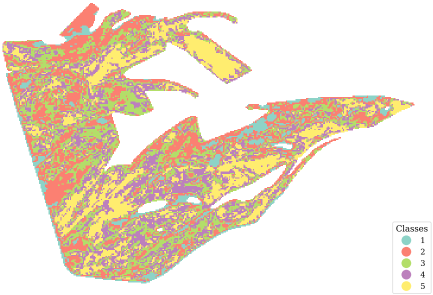
The Workflow:

1. **Identify isolated spots:** Use math to flag pixels that stand out from their neighbors.
2. **Model the expected value:** Use Kriging to predict what the physical value should be at those locations.
3. **Test the consistency:** Check if the current labels actually match the underlying physical data.
4. **Reassign labels:** Use likelihood-based math to correct any labels that don't fit in.

Data Overview: Spatial Distribution



(a) Productivity Distribution (Z)



(b) Fragmented Categorical Map (L)

Here, the continuous physical attribute Z is the productivity index, and the categorical map L is the result of EM clustering.

- The continuous physical field (a) shows attribute values range from 0.39 to 0.92, and exhibits a spatial gradient with clear spatial autocorrelation.
- The categorical map (b) exhibits high-frequency noise.

Verification of Prerequisite: Spatial Autocorrelation

Fundamental Assumption

Ordinary Kriging requires the attribute field $Z(\mathbf{X})$ to exhibit spatial autocorrelation, rather than complete spatial randomness [2].

Quantitative Assessment (Global Moran's I)

$$I = \frac{N}{W} \frac{\sum_i \sum_j w_{ij}(Z_i - \bar{Z})(Z_j - \bar{Z})}{\sum_i (Z_i - \bar{Z})^2} \tag{1}$$

- **Result:** A highly significant positive spatial autocorrelation ($I = 0.8002$, $p < 0.001$).
- **Implication:** This provides the theoretical support to model the physical field $Z(\mathbf{X})$ as a **Spatial Random Field (SRF)** and apply Kriging methods.

Data Normalization

Normality Violation: Original data is apparently non-Normal distributed. This violates the normality assumption in later hypothesis testing, which will lead to biased predictions and invalid hypothesis testing [5].

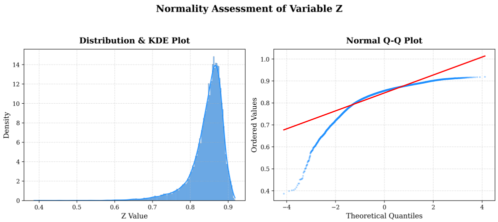
Normal Score Transformation (NST)

$$Y(\mathbf{X}_i) = \Phi^{-1}(F(Z(\mathbf{X}_i))) \qquad (2)$$

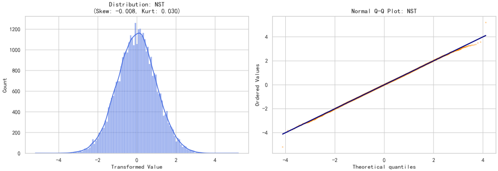
** Φ^{-1}: Inverse standard normal CDF; F: Cumulative Distribution Function of Z.*

Transformation Results:

- Regularized spatial skewness to ≈ 0.
- The empirical quantiles align with the Gaussian diagonal (see Q-Q plot).



(a) Original Z Histogram & Q-Q Plot



(b) NST Transformed Y Histogram & Q-Q Plot

Step 1: Isolation Detection (Candidate Extraction)

Objective: Identify spatially isolated pixels, which are likely misclassified rather than being true geographical features.

Mathematical Metrics

- **Neighbor Disagreement Score (NDS):** Quantifies spatial contradiction.

$$NDS(\mathbf{X}_i) = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} \mathbb{I}\Big(L(\mathbf{X}_j) \neq L(\mathbf{X}_i)\Big) \tag{3}$$

  * $\mathbb{I}(\cdot)$ *is the indicator function;* $\mathcal{N}_i$ *defines the Moore neighborhood.*

- **Connected Components Analysis (CCA):** Evaluates topological connectivity to flag clusters below the designated Minimum Unit.

Algorithmic Logic: Denote $\mathcal{S}$ as the set of all pixels, $\mathcal{S}_{\text{NDS}}$ as the set of pixels with $NDS > \tau$ (e.g., $\tau = 0.8$), and $\mathcal{S}_{\text{CCA}}$ as the set of pixels in connected components smaller than the Minimum Unit (e.g., 4 pixels). The pixels within the union set $\mathcal{S}_{\text{cand}} = \mathcal{S}_{\text{NDS}} \cup \mathcal{S}_{\text{CCA}}$ are considered isolated and serve as prediction targets in the subsequent Kriging stage.

Step 2: Geostatistical Validation (Ordinary Kriging)

For candidate $\mathbf{X}_0 \in \mathcal{S}_{\text{cand}}$, we use Ordinary Kriging to predict the expected value $\hat{Z}(\mathbf{X}_0)$ and quantify the associated uncertainty $\sigma_K^2(\mathbf{X}_0)$.

Kriging Methodology

$$\textbf{Estimator:}\ \hat{Z}(\mathbf{X}_0) = \sum_{i=1}^{n} \lambda_i Z(\mathbf{X}_i), \quad \mathbf{X_i} \in \mathcal{S}_{\text{reference}} = \mathcal{S}_{\text{cand}} \backslash S \tag{4}$$

$$\textbf{Variance:}\ \sigma_K^2(\mathbf{X}_0) = 2\sum_{i=1}^{n} \lambda_i \gamma(\mathbf{X}_i, \mathbf{X}_0) - \sum_{i=1}^{n}\sum_{j=1}^{n} \lambda_i \lambda_j \gamma(\mathbf{X}_i, \mathbf{X}_j) \tag{5}$$

λ_i are the Kriging weights determined by solving the Kriging system of equations, γ is the semivariogram to model spatial dependence.

Why Ordinary Kriging?

- Best Linear Unbiased Predictor.
- Provides physically grounded uncertainty σ_K^2.

Computational Optimization: The computation complexity is $\mathcal{O}(N^3)$, therefore utilizes only reference pixels that are within a certain distance from the candidate pixel.

Step 3: Hypothesis Testing Framework

- H_0 **(The "Normal" case):** The observed value at this location is consistent with the surrounding area. It looks like something that would naturally occur given its neighbors.
- H_A **(The "Outlier" case):** The observed value is out of place. It seems to come from a different process, meaning it doesn't fit in with the nearby environment.

Test Statistic $\eta(\mathbf{X}_0)$:

$$\eta(\mathbf{X}_0) = \frac{Z_{\text{obs}}(\mathbf{X}_0) - \hat{Z}(\mathbf{X}_0)}{\sigma(\mathbf{X}_0)} \sim \mathcal{N}(0, 1) \tag{6}$$

** Asymptotic normality is mathematically guaranteed by the prior NST step.*

To avoid the multiple testing problem across all candidates, we apply the Benjamini-Hochberg procedure to control the False Discovery Rate (FDR) at a pre-specified level $q = 0.05$.

Decision Rule

1. **Reject H_0:** Large physical discrepancy $\rightarrow$ **Preserve original label** (Real feature).
2. **Fail to reject H_0:** Small discrepancy $\rightarrow$ **Proceed to Reassignment** (Noise).

Step 4: Likelihood-Based Class Reassignment

Kriging-Based Likelihood Function

For each candidate class $C_k \in \mathcal{C}$, we compute the likelihood of observing $Z_{\text{obs}}(\mathbf{X}_0)$:

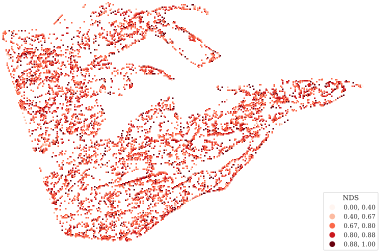
$$\mathcal{L}(C_k, \mathbf{X}_0) = \frac{1}{\sigma_{C_k}(\mathbf{X}_0)} \exp\left(-\frac{(Z_{\text{obs}}(\mathbf{X}_0) - \hat{Z}_{C_k}(\mathbf{X}_0))^2}{2\sigma^2_{C_k}(\mathbf{X}_0)} \right) \tag{7}$$

** $\hat{Z}_{C_k}(\mathbf{X}_0)$ and $\sigma^2_{C_k}(\mathbf{X}_0)$ are the Kriging prediction and variance computed using only the reference pixels belonging to class C_k, i.e., $\mathcal{S}_{\text{ref}}(C_k) = \{\mathbf{X}_i : L(\mathbf{X}_i) = C_k\}$.*

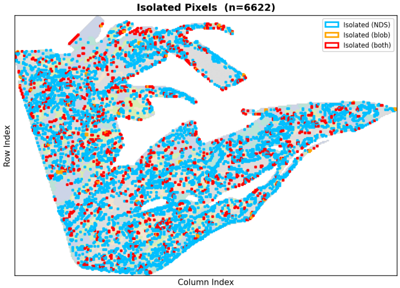
- **Structural Regularization:** The leading term $\sigma^{-1}_{C_k}$ mathematically penalizes candidate classes that exhibit high spatial uncertainty or sparse local support.
- The corrected final label L^* is chosen to maximize the likelihood function across all candidate classes $\mathcal{C}$ (i.e., all 5 classes):

$$L^*(\mathbf{X}_0) = \arg\max_{C_k \in \mathcal{C}} \mathcal{L}(C_k, \mathbf{X}_0)$$

Result 1: Extraction of Candidate Outliers



(a) Continuous NDS Distribution



(b) Extracted Spatial Candidates

Observations from Step 1:

- **NDS Map (a):** High spatial contradictions are primarily clustered along class boundaries.
- **Candidates (b):** Effectively isolates the "salt-and-pepper" noise patches. **6,622 pixels** flagged as candidates ($\approx 18.4\%$ of total domain).

Result 2: Semivariogram Modeling

Based on the weighted Root Mean Square Error (wRMSE), the Exponential model demonstrated the most optimal fit.

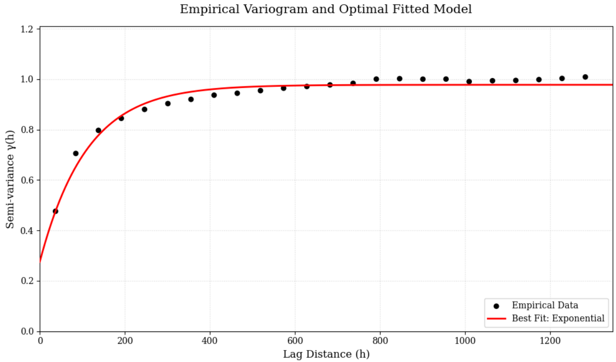
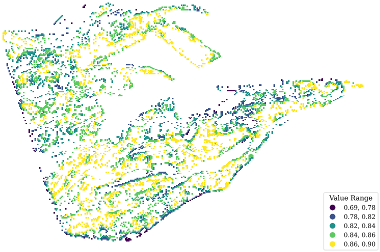


Figure 4: Empirical Semivariogram (black dots) and Fitted Models: Exponential (red).

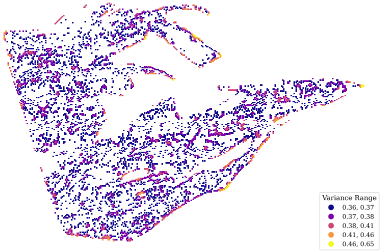
Fitted Parameters:

- **Nugget (c_0):** 0.275, reflects inherent sensor/environmental noise.
- **Sill (c):** 0.978, indicates the total variance of the spatial process.
- **Range (a):** 110.0 meters, defines the spatial correlation distance.

Result 3: Expected Realities and Uncertainty



(a) Kriging Prediction Field ($\hat{Z}$)



(b) Kriging Variance Field (σ_K^2)

- **Prediction (a):** Successfully maps the underlying smooth continuous field behind the categorical noise.
- **Variance (b):** Uncertainty is spatially quantified ($\sigma_K^2 \in [0.36, 0.65]$). High variance zones correspond to areas with sparse data support, guiding the subsequent hypothesis testing.

Result 4: Statistical Decisions

False Discovery Rate (FDR) Control at $q = 0.05$:

Statistical Category	Pixel Count	Percentage
Total Isolated Candidates (S_{cand})	6,622	100%
Rejected H_0 (Genuine Anomalies)	**6**	**0.1%**
Fail to reject H_0 (Artifacts/Noise)	6,616	99.9%

Interpretation & Impact:

- The visual "salt-and-pepper" noise is overwhelmingly confirmed as non-physical artifacts (99.9%).
- Crucially, the conservative FDR test successfully identifies and **preserves 6 legitimate micro-structures** that would have been destroyed by standard filters.

Result 5: Label Correction Analysis (MLE)

MLE Reassignment Summary:

- **Reassigned (5,180):** Transitioned to a new label that maximizes the physical likelihood $\mathcal{L}(C_k, \mathbf{X}_0)$.
- **Retained (1,436):** Geometrically isolated, yet **statistically most probable** in their original class.

The Defeat of Heuristic Filters

This proves that a pure "Majority Voting" algorithm would have mistakenly flipped the labels of **21.7%** (1,436/6,616) of the isolated pixels.

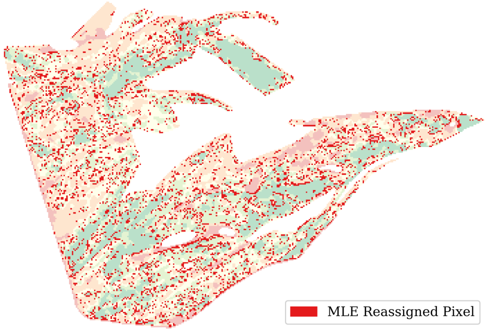
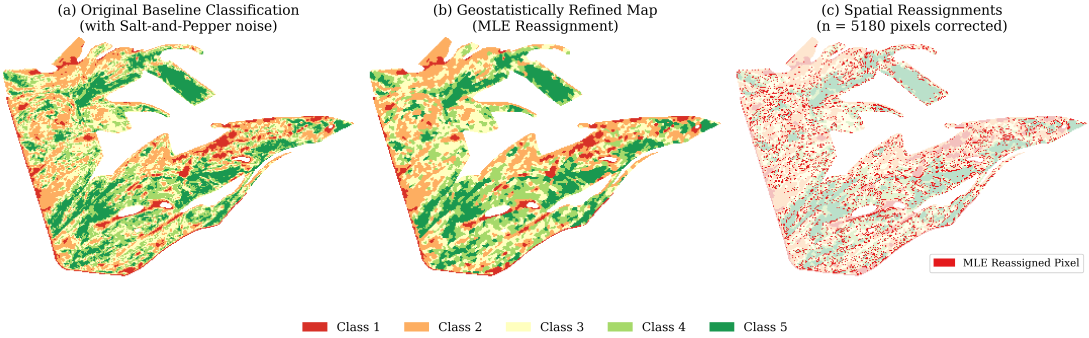


Figure 6: Final Refined Categorical Map.

Comprehensive Map Comparison



(a) Original Fragmented **(b) Refined Map** **(c) Difference Map**

- **Visual Assessment:** The refinement process (b) successfully eliminated spurious noise while strictly preserving clear linear boundaries and true geographic structure.

Discussion and Methodological Limitations

1. Theoretical and Empirical Discussion

- **High reliability:** We rejected the null hypothesis in only 0.1% of cases. This shows that 99.9% of the isolated areas are actually consistent with their surrounding environment.
- **Validating isolated labels:** Our analysis kept 21.7% (1,436 pixels) of the geometrically isolated labels. This proves that just because a label is isolated in space doesn't automatically mean it's an error.
- **Accounting for uncertainty:** By using local variance ($\sigma_{C_k}^{-1}$) as a weight, the model naturally gives less importance to predictions that are either too scattered or lack enough nearby data, making the results more robust.

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2. Methodological Limitations

- **Assumption of constant average:** We assume that the data's average value stays roughly the same across the area. If there are clear trends—like a steady change in elevation—this assumption breaks down. This leads to biased results and increases the risk of making wrong conclusions.
- **Slow performance:** Unlike simpler methods that process data quickly, this approach requires heavy matrix math. Because the work required grows rapidly as you add more neighbors ($O(m^3)$ complexity), it becomes very slow and difficult to use on massive datasets.

Conclusion & Future Directions

Research Summary:

- Transitioned post-classification from "geometric cleanup" to **formal spatial inference**.
- Successfully balanced aggressive noise eradication with the preservation of micro-scale geographic realities.

Future Research Directions:

- Replace OK with **Universal Kriging** or **Regression Kriging** to incorporate auxiliary variables (e.g., DEM, Soil Moisture).
- Expanding $\gamma(\mathbf{h})$ into **3D semivariograms** $\gamma(\mathbf{h}, t)$ to validate pixels against their historical physical trajectories.
- Utilizing cross-correlation between multiple spectral indices to enhance refinement stability.

References I

[1] Julian Besag. "On the statistical analysis of dirty pictures". In: *Journal of the Royal Statistical Society: Series B (Methodological)* 48.3 (1986), pp. 259–279.

[2] Noel Cressie. *Statistics for spatial data*. John Wiley & Sons, 2015.

[3] Kwang E Kim. "Adaptive majority filtering for contextual classification of remote sensing data". In: *International Journal of Remote Sensing* 17.5 (1996), pp. 1083–1087.

[4] Waldo R Tobler. "A computer movie simulating urban growth in the Detroit region". In: *Economic geography* 46.2 (1970), pp. 234–240.

[5] Hans Wackernagel. *Multivariate geostatistics: an introduction with applications*. Springer Science & Business Media, 2003.

Thank You!

Questions & Discussion

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